

ALEXANDER NICKITA

Software Engineer | Systems & Embedded

Software engineer with 2+ years building and shipping software applications in C, C++, Python, and Java across embedded systems and Linux environments. Currently delivering defense-grade firmware at General Dynamics Land Systems, with a track record of code reviews, cross-functional collaboration, and scalable solutions validated in Agile sprints. Dean's List every semester at LTU (3.62 GPA), SAFe 6 Scrum Master certified, and IEEE Chapter Chair — bringing both the technical depth and best practices discipline to grow fast on a software engineering team.

EMPLOYMENT HISTORY

EMBEDDED LINUX SOFTWARE ENGINEER (INTERN) General Dynamics Land Systems

May 2026 – Aug 2026
Sterling Heights, Michigan

Developed and maintained scalable software applications for defense-grade ground vehicle systems, measured by successful subsystem-level firmware validation, by writing C software in an Embedded Linux environment and implementing Yocto build recipes following team code review and best practices standards.

Improved hardware-software integration quality, measured by zero integration defects at handoff, by collaborating daily with a cross-functional engineering team of senior engineers to design, implement, and review new features against real-time system requirements in an Agile workflow.

Accelerated board bring-up timelines, measured by meeting sprint delivery milestones, by authoring Python and C/C++ toolchain scripting for Yocto-based Linux image builds, applying industry best practices for embedded software development.

RESEARCH ASSISTANT Lawrence Technological University

Aug 2024 – Present
Southfield, Michigan, United States

Accelerated embedded sensor pipeline development, measured by a fully functional Raspberry Pi 5 presence-detection prototype scoped under a university research budget, by leading hardware selection, designing the MobileNet transfer-learning architecture, and implementing Python data collection software following best practices for reproducibility.

Supported a VR-based cognitive assessment study (Corsi Block Test) for brain trauma research, measured by successful participant data collection across multiple cohorts, by proctoring sessions, maintaining experimental protocols, and onboarding undergraduate participants against methodology standards.

COMPUTER ASSISTANT Southfield Public Library

Nov 2024 – Present
Southfield, Michigan, United States

Improved patron technology access, measured by consistent resolution of walk-in software and hardware issues, by troubleshooting workstations, assisting users with productivity applications, and applying best practices for public-facing IT support.

EDUCATION

BACHELOR OF APPLIED SCIENCE - BASc IN EMBEDDED SOFTWARE ENGINEERING Lawrence Technological University

Aug 2024 – Present
Southfield, Michigan, United States

Expected Graduation: May 2027 GPA: 3.62

ASSOCIATE'S DEGREE IN COMPUTER SOFTWARE ENGINEERING Oakland Community College

Jan 2022 – May 2024
United States

Graduated with honors, 3.75 GPA, Magna Cum Laude

SKILLS

Languages: C/C++, Python, Java | Frameworks & Tools: Embedded Linux, Yocto Project, OpenCV, TensorFlow Lite | Protocols: UART / SPI / I2C, Bluetooth / BLE, MQTT | Platforms: Raspberry Pi, Linux, Microcontroller Programming (TM4C123G, ESP32) | Practices: Agile, SAFe 6 Scrum Master (Certified), Code Review, UML Modeling, Data Structures, Digital Logic Design

ADDITIONAL INFORMATION

ENGINEERING PROJECTS

Bare-Metal Smart Home Controller (TM4C123G)

Designed and implemented a bare-metal IoT smart home controller on the TM4C123G microcontroller using C. Integrated Bluetooth (HC-05), DHT11 temperature/humidity sensing, WS2812B addressable LED control, PIR motion detection, and fan speed control via PWM. Developed custom UART, SPI, and GPIO drivers from scratch with no OS or HAL, producing full technical documentation and demo materials.

Edge Vision Node — Embedded Smart Home Sensing Module

Developing a low-cost embedded IoT sensing module on Raspberry Pi 5 with a camera module, environmental sensors (BME280, PIR), and ESP32 nodes. Uses OpenCV and TensorFlow Lite for on-device real-time human presence detection and activity classification, communicating with a smart home control system via standard IoT protocols to trigger automated actions without cloud dependency.

ACADEMIC PROJECTS

Computer Vision-based Identification Application

Built a real-time computer vision application in C on a Tiva C-series microcontroller, training an OpenCV-based AI model to classify humans vs. objects with reliable detection accuracy. Delivered end-to-end from model training through hardware integration and live demo.

AWARDS

LAWRENCE TECHNOLOGICAL UNIVERSITY

Outstanding Member of a Student Organization (IEEE)

Elected Chairman of LTU's IEEE Student Branch; recognized by faculty as Outstanding Member for driving chapter growth, organizing technical events, and representing the branch at university level.

LAWRENCE TECHNOLOGICAL UNIVERSITY

Virinder K. Moudgil Academic Merit Scholarship

Competitive academic merit scholarship awarded to top-performing transfer students entering LTU's engineering program; selected on the basis of GPA (3.75, Magna Cum Laude at OCC) and academic trajectory.

LAWRENCE TECHNOLOGICAL UNIVERSITY

Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring, 2026)

Maintained 3.50+ GPA every semester at LTU while simultaneously interning at General Dynamics and serving as IEEE Chapter Chair.